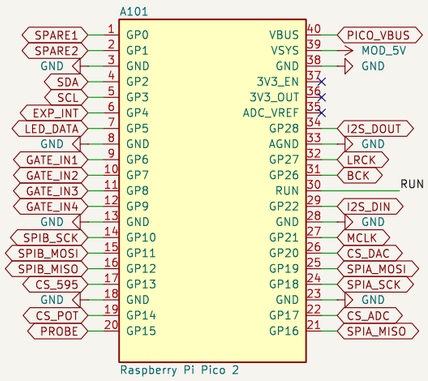
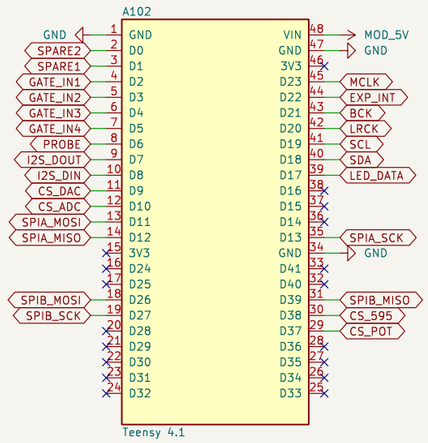
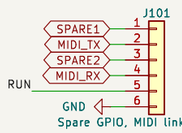
ANALOGUE BOARD REV A — COMPUTE MODULES AND PIN MAP

Both footprints sit on the one board, wired to the same nets; only one module is fitted.
Every signal into a module is 3.3 V: the Teensy 4.1 is not 5 V tolerant.
Pico 2: 5 V reaches VSYS through D203 (Schottky, net MOD_5V), VBUS is left to USB.
Teensy 4.1: cut the VIN-VUSB pads before fitting; VIN takes MOD_5V.
Neither module's own 3.3 V output is used: the board makes its own +3V3A and +3V3D.
ESP32-P4-WIFI6 (Pico pinout): firmware must set LDO channel 4 to 3.3 V first (pins 21, 22, 24 are on that bank); bridge JP201 for its supply margin. See docs/analogue_board_esp32p4.md.







MIDI: bridge SPARE1–MIDI_TX and SPARE2–MIDI_RX (adjacent holes) with a wire or shunt. Left open, SPARE1/2 are plain GPIO.
On a Pico, move the SDK console off UART0 to USB.
RUN: wire a reset button to GND (Pico or ESP32-P4).

PIN MAP

Net	Function	Pico 2	Teensy 4.1	Why this pin
MCLK	I2S master clock to SCKI1/2	GP21 (pin 27)	23	Pico CLK_GPOUT0 · Teensy MCLK1
BCK	I2S bit clock	GP26 (pin 31)	21	Pico PIO side-set · Teensy BCLK1
LRCK	I2S word clock	GP27 (pin 32)	20	Pico PIO side-set +1 · Teensy LRCLK1
I2S_DOUT	I2S data to codec DIN	GP28 (pin 34)	7	Teensy OUT1A
I2S_DIN	I2S data from codec DOUT	GP22 (pin 29)	8	Teensy IN1
SPIA_SCK	Converter bus clock	GP18 (pin 24)	13	Pico SPI0 SCK · Teensy SCK
SPIA_MOSI	Converter bus data out	GP19 (pin 25)	11	Pico SPI0 TX · Teensy MOSI
SPIA_MISO	Converter bus data in (ADS8688)	GP16 (pin 21)	12	Pico SPI0 RX · Teensy MISO
CS_ADC	ADS8688 chip select	GP17 (pin 22)	10	Pico SPI0 CSn · Teensy CS
CS_DAC	DAC8568 SYNC	GP20 (pin 26)	9	
SPIB_SCK	Panel bus clock	GP10 (pin 14)	27	Pico SPI1 SCK · Teensy SCK1
SPIB_MOSI	Panel bus data out	GP11 (pin 15)	26	Pico SPI1 TX · Teensy MOSI1
SPIB_MISO	Panel bus data in (MCP3208)	GP12 (pin 16)	39	Pico SPI1 RX · Teensy MISO1 (alt pin)
CS_595	74HCT595 latch (RCLK)	GP13 (pin 17)	38	
CS_POT	MCP3208 chip select	GP14 (pin 19)	37	
SDA	I2C data	GP2 (pin 4)	18	Pico I2C1 SDA · Teensy SDA (Wire)
SCL	I2C clock	GP3 (pin 5)	19	Pico I2C1 SCL · Teensy SCL (Wire)
EXP_INT	TCA9555 interrupt, active low	GP4 (pin 6)	22	
GATE_IN1	Gate input 1	GP6 (pin 9)	2	
GATE_IN2	Gate input 2	GP7 (pin 10)	3	
GATE_IN3	Gate input 3	GP8 (pin 11)	4	
GATE_IN4	Gate input 4	GP9 (pin 12)	5	
PROBE	Jack-detect probe	GP15 (pin 20)	6	
LED_DATA	WS2812 data (buffered to 5 V)	GP5 (pin 7)	17	Teensy TX4 (WS2812Serial)
SPARE1	Spare GPIO 1, or MIDI out (link hole)	GP0 (pin 1)	1	Pico UART0 TX · Teensy TX1
SPARE2	Spare GPIO 2, or MIDI in (link hole)	GP1 (pin 2)	0	Pico UART0 RX · Teensy RX1
MOD_5V	Module 5 V in	VSYS (pin 39)	VIN	Schottky D203; Q201 bypass if JP201
PICO_VBUS	Module USB 5 V sense	VBUS (pin 40)	–	Q201 gate: off with USB plugged in
RUN	Pico reset	RUN (pin 30)	–	To hole J101-5

SHEETS

Power
Power entry, buck, linear rails, test points
File: power.kicad_sch

Codec & audio
PCM3060, two audio ins, two audio outs
File: codec_audio.kicad_sch

CV jacks & ADC
ADS8688, DG412 jack switches, CV holes
File: cv_io.kicad_sch

CV DAC
DAC8568 and the eight ±5 V output stages
File: cv_dac.kicad_sch

Pots & gate outs
MCP3208 pots, 74HCT595 gate outputs
File: pots_gates.kicad_sch

Expander, I2C & LEDs
TCA9555 buttons, EEPROM, Qwiic, WS2812
File: expander_i2c.kicad_sch

Gate ins & jack detect
LM393 gate inputs, probe resistors
File: gate_in.kicad_sch

Edge strips
n8synth edge strips, fiducials
File: edge_strips.kicad_sch

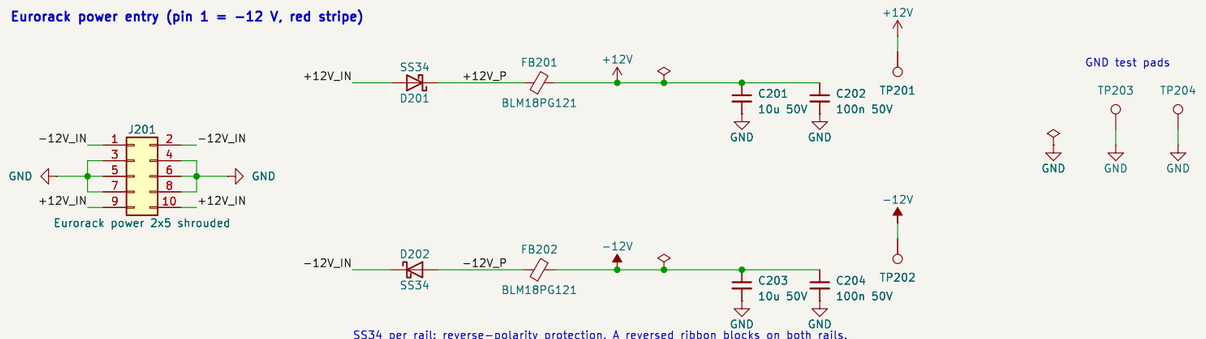
TRS MIDI
MIDI in (H11L1) and out (5 V buffered), Type A
File: midi.kicad_sch

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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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Sheet: /
File: analogue_board.kicad_sch
Title: Compute modules and pin map
Size: A3 | Date: 2026-10-14 | **Rev: A**
KiCad E.D.A. 9.0.7 | Id: 1/10

POWER

±12 V enter through a 2×5 shrouded header (the bus +5 V is not used). Each rail: series Schottky, ferrite, 10 µF + 100 nF. +5VD comes from a forced-PWM buck (no light-load pulse skipping in the audio band); +5VA from a linear regulator on its own; each 3.3 V rail from its own LDO. A 0 Ω link after every regulator lets a faulty rail be isolated and its current measured.

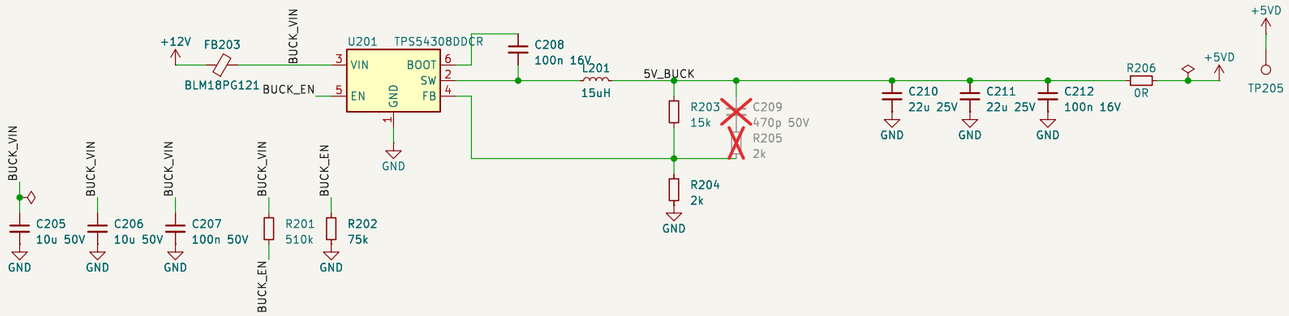
Eurorack power entry (pin 1 = −12 V, red stripe)



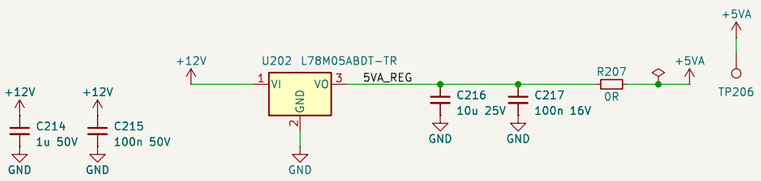
SS34 per rail: reverse-polarity protection. A reversed ribbon blocks on both rails. 3 A parts: +12 V can reach ≈0.75 A with an ESP32-P4 and a lit LED chain.

+5VD: TPS54308 forced-PWM buck, 350 kHz (TPS54202 is a pin- and value-compatible fallback)

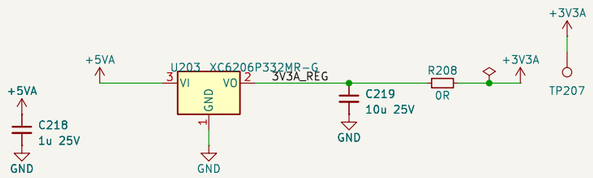
Vout = 0.596 V × (1 + 15k/2k) = 5.07 V; L and Cout from TI's table for 5 V. Input: own ferrite and 2 × 10 µF, keeping the 350 kHz input ripple off the ±12 V rail the op-amps share. EN divider 510k/75k: starts at ≈9.2 V, stops at ≈8.1 V, so the digital 5 V never runs while the analogue 5 V (L78M05) is down: TI's power-up order for the codec.



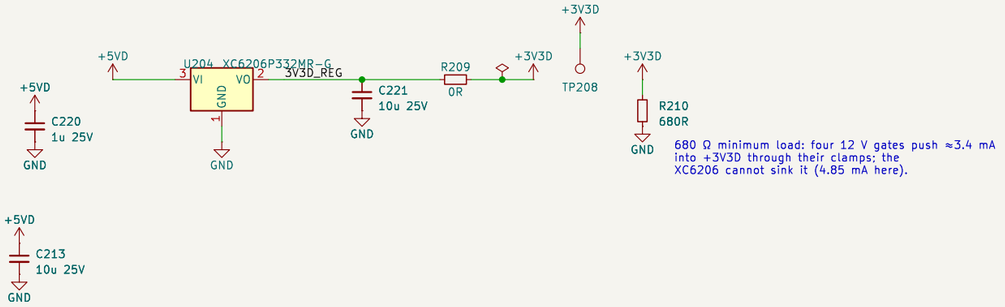
+5VA: L78M05 from +12 V, its own regulator (PCM3060 VCC, ADS8688 AVDD, +3V3A LDO)



+3V3A: XC6206 from +5VA (DAC8568, MCP3208, pot reference via POT_REF)

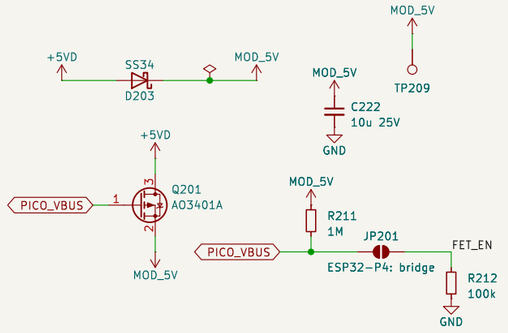


+3V3D: XC6206 from +5VD (codec VDD, ADS8688 DVDD, logic, Qwiic, MIDI)



MOD_5V: +5VD through a Schottky to Pico VSYS / Teensy VIN / ESP32-P4 VCC_5V

Q201 bypasses D203 when JP201 is bridged, for the ESP32-P4, whose regulator needs ≥ 4.5 V at 0.5–0.7 A. Its gate follows the Pico/P4 VBUS pin (pin 40): with USB plugged in the FET turns off, as in the Pico 2 datasheet's VSYS circuit. Leave JP201 open for a Teensy (an uncut VIN–VUSB link would tie USB to the buck).



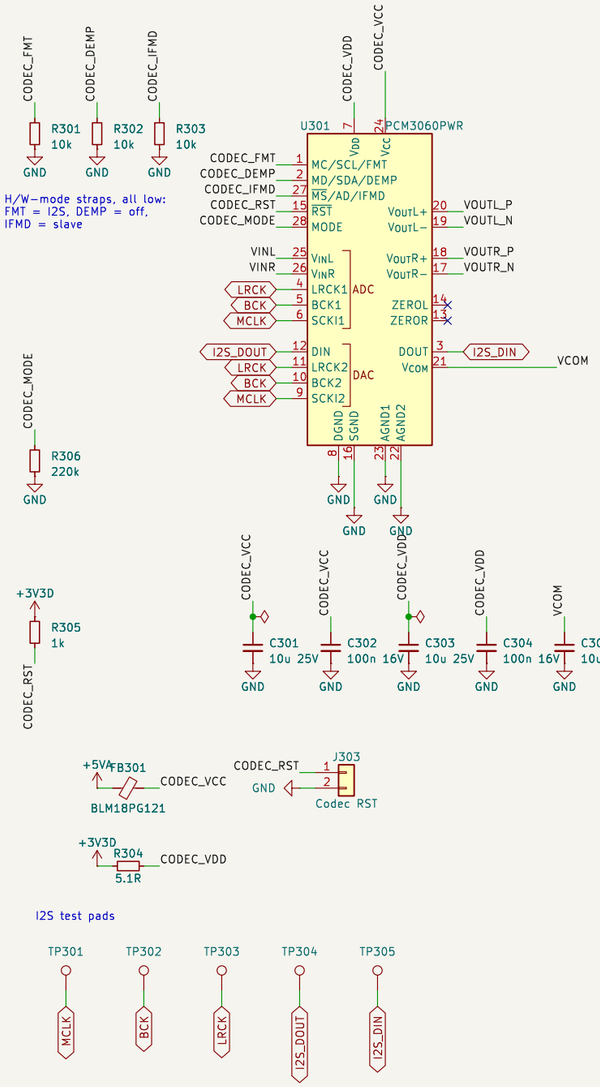
Load estimate: +12 V ≈ 120 mA typical, ~350 mA worst case with a Teensy and a lit LED chain; with an ESP32-P4 (0.45–0.7 A at 5 V) plus LEDs, ≈0.75 A. −12 V ≈ 25 mA. XC6206: 200 mA each; +3V3D ≈ 50 mA.

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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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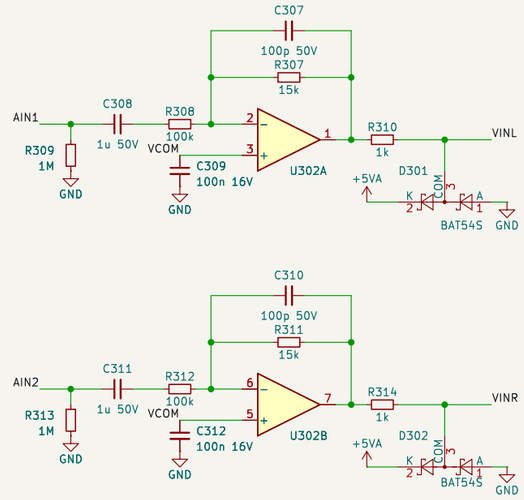
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File: power.kicad_sch		
Title: Power		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 2/10

CODEC AND AUDIO

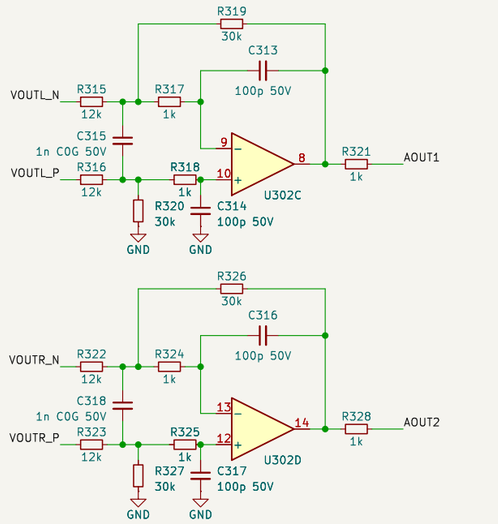
PCM3060 in hardware-control mode, the same straps as Mutable Beads: MODE 220 kΩ to DGND = parallel control with differential outputs; FMT low = 24-bit I2S, IFMD low = ADC and DAC both slaves, DEMP low = no de-emphasis. One I2S bus feeds both halves. RST is pulled up, so the codec starts by itself; CODEC_RST also reaches a hole (J303) so firmware can hold it in reset, as Beads does. Full scale is 20 Vpp in and out; firmware runs about 6 dB below. VCC via a ferrite from +5VA, VDD via 5.1 Ω from +3V3D: each with its own 10 µF + 100 nF.

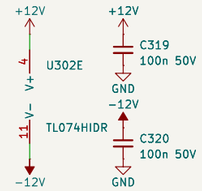


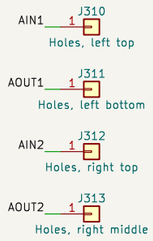
AUDIO INPUTS: 22 Vpp → 3 Vpp (codec full scale). AC-coupled, inverting (firmware flips the sign). ±12 V stays inside the codec rails, so the BAT54S clamps only act on faults. 100 nF on each + input decouples VCOM (12.5 kΩ source) at the op-amp, as the Patch SM does.



AUDIO OUTPUTS: balanced second-order MFB low-pass (Beads' topology), gain 30k/12k = 2.5, so 8 Vpp → 20 Vpp, DC-coupled, non-inverting (VOUT+ to the + side). 1 nF C0G across the two nodes and 100 pF from each op-amp input: f0 ≈ 65 kHz, Q ≈ 0.73, flat to 20 kHz (Basic parts except the 1 nF C0G). 1 kΩ 1206 out survives output-to-output patching.







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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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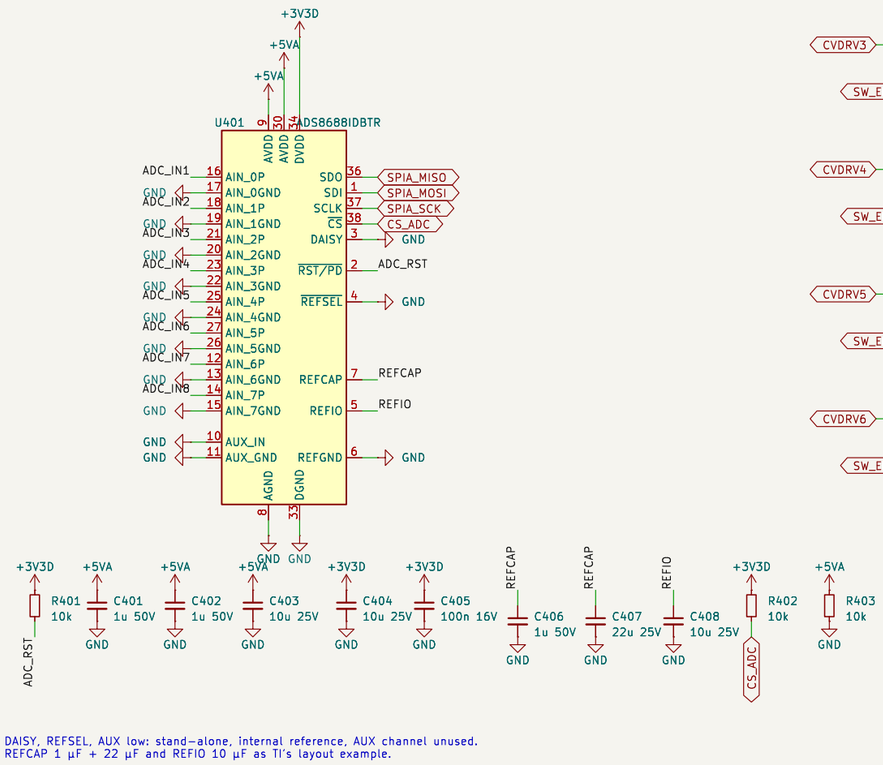
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File: codec_audio.kicad_sch

Title: Codec & audio

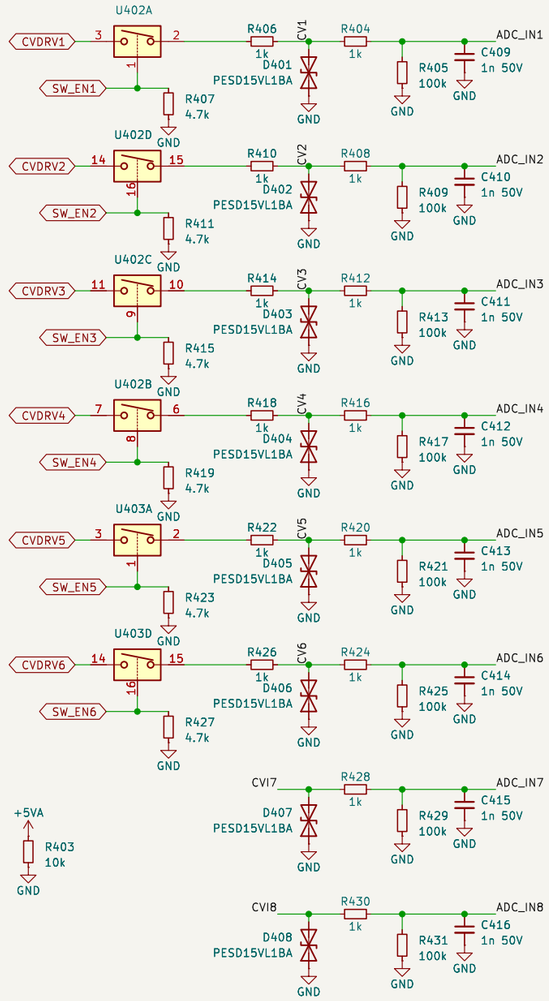
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KiCad E.D.A. 9.0.7		Id: 3/10

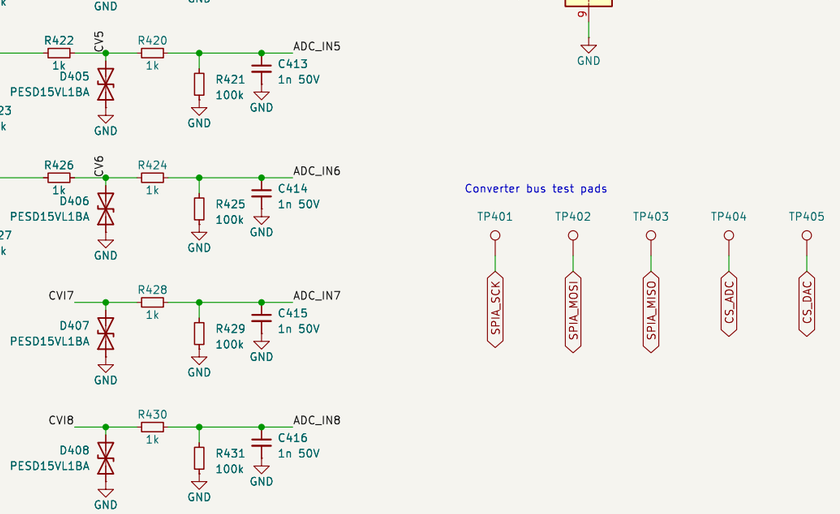
CV JACKS AND ADC

ADS8688: 8 channels, 16-bit, 1 MΩ inputs with ±10.24 / ±5.12 V ranges per channel (firmware default ±10.24 V), OVP ±20 V.
Each CV jack: 15 V TVS at the hole, read through 1 kΩ with 1 nF at the ADC pin (TI's EVM). 100 kΩ to ground, but the ADC's 1 MΩ returns to an internal bias, so an UNPATCHED input reads ≈ +0.2 V: firmware trusts the jack-detect probe.
CV1-6 are bidirectional: a DG412 switch (on at logic 1) joins the DAC stage through 1 kΩ (1206); with it on the ADC reads the jack itself. 4.7 kΩ pull-downs beat the TCA9555's internal pull-ups: every jack starts as an input.

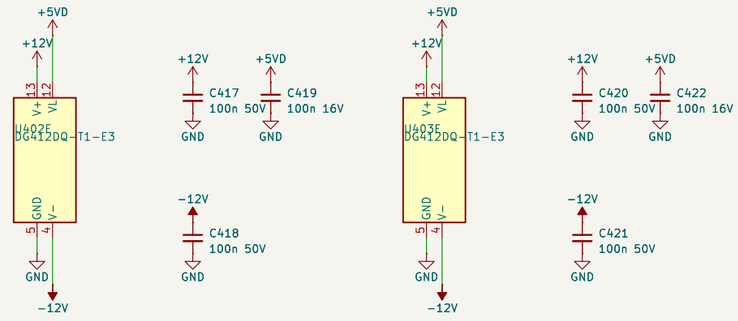


DAISY, REFSEL, AUX low: stand-alone, internal reference, AUX channel unused.
REFCAP 1 µF + 22 µF and REFIO 10 µF as TI's layout example.





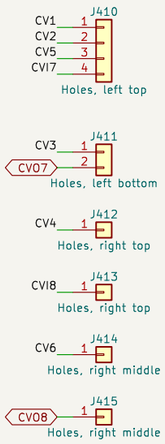
CV07, CV08 (out only): their 1 kΩ sits inside the DAC stage's feedback loop on the CV DAC sheet, so the jack holds its voltage under load. Calibrate them with a cable to CVI7 / CVI8.



Spare switches (U403 SW3, SW4): off, both sides grounded



Converter bus test pads



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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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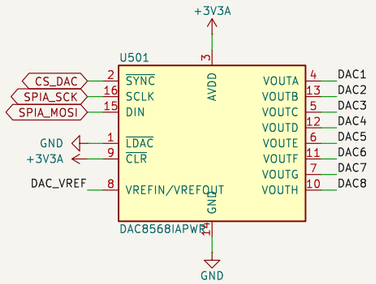
Sheet: /CV jacks & ADC/
File: cv_io.kicad_sch

Title: CV jacks & ADC

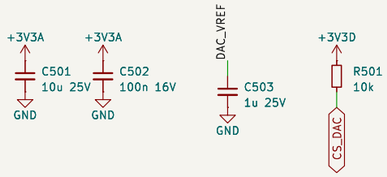
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 4/10

CV DAC

DAC8568 grade A: 0–2.5 V from its internal reference (off at power-up), outputs reset to zero. Each stage: non-inverting gain 1 + 33k/10k = 4.3, offset by OFFSET_REF = VREF × 22k/34k through the 10 kΩ: Vout = 4.3·Vdac − 5.34 V, so −5.34 V … +5.41 V, putting ±5 V inside the DAC's specified codes (0 V at code ≈ 32544). The offset tracks the DAC's own reference, as in Performer, O&C, Marbles and Stages. CVO7/8: the 1 kΩ sits inside the loop (feedback from the jack side, as Mutable does), so the jack holds its voltage under load. CV1–6 go through the DG412 switch; firmware corrects their load sag using the ADC, which reads the jack. ±10 V variant (not Rev A): Rf 75k, divider 15k/20k, all Basic: gain 8.5, OFFSET_REF 1.429 V, −10.71 … +10.54 V; needs ≥ 10 kΩ loads. The TL074H can swing it.

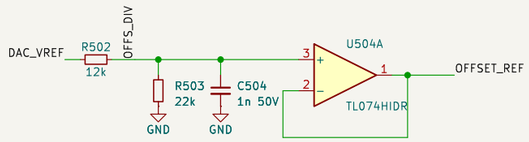


LDAC low: each write updates its output.
CLR high: clear pin unused.

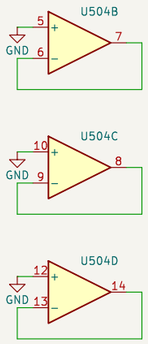


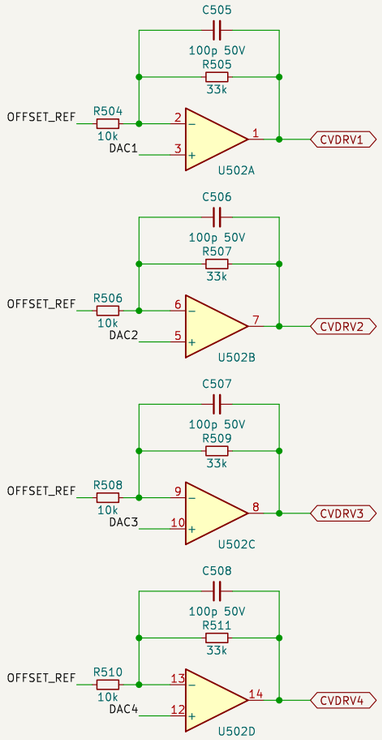
OFFSET_REF = VREF × 22k/(12k + 22k) = 1.618 V, buffered

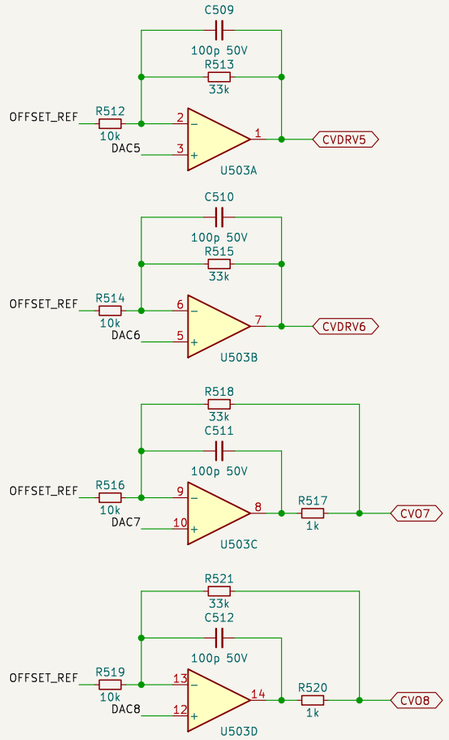
1 nF (not 1 µF): the offset must track VREF quickly, or enabling the reference glitches every output.



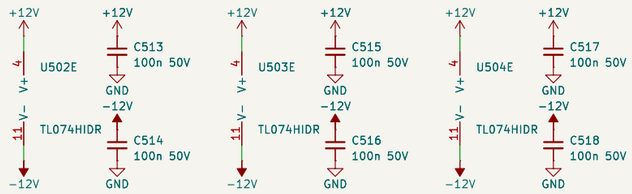
Spare U504 sections: followers with + to GND







CVDRV1–6 go to the DG412 switches (sheet CV jacks & ADC). CVO7–8 go straight to their holes.



Generated by tools/analogue_board/build.py: edit the generator, not these files
Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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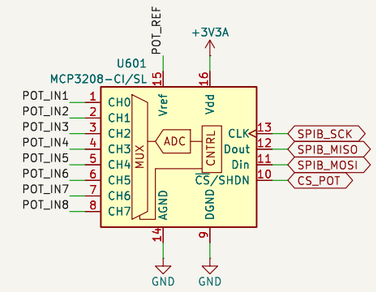
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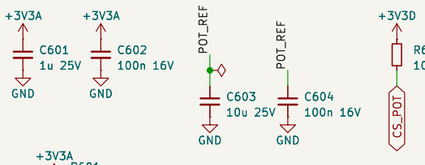
Title: CV DAC

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 5/10

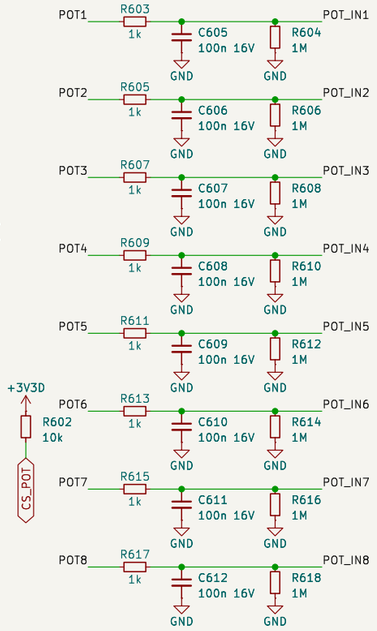
POTS AND GATE OUTPUTS

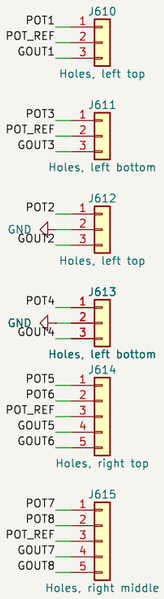
Pots: MCP3208 on +3V3A. Its VREF and the pot tops share POT_REF (100 Ω + 10 µF from +3V3A), so readings are ratiometric and the hand-wired pot leads can't disturb the DAC's supply. Each wiper: 1 kΩ + 100 nF, 1 MΩ to ground so an unwired channel reads 0 (100 kΩ would bend a 100 kΩ pot's response by up to 12 %).
Gate outs: 74HCT595 at 5 V (VIH 2.0 V, so 3.3 V logic drives it) → 1 kΩ → jack, BAV99 clamps to +5VD and GND.
OE is held high (outputs off) by a pull-up until firmware drives GATE_OE_N low: the 595's register powers up random.

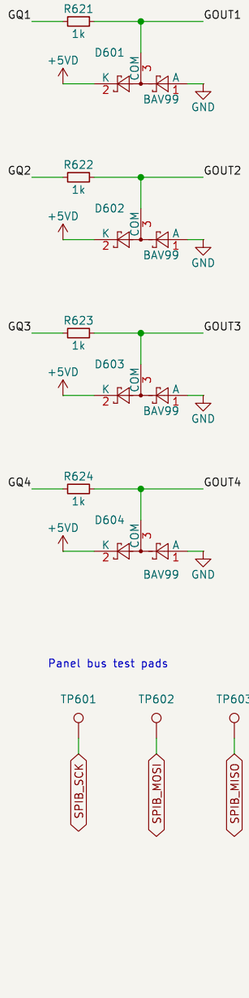


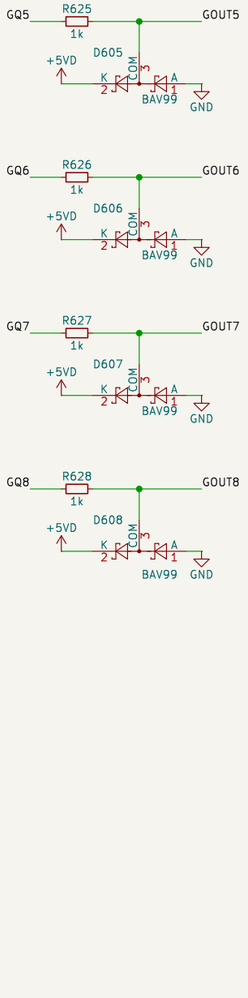




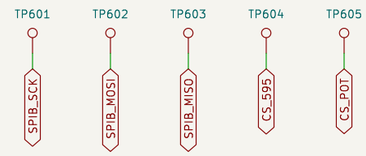


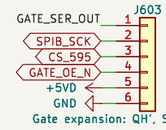






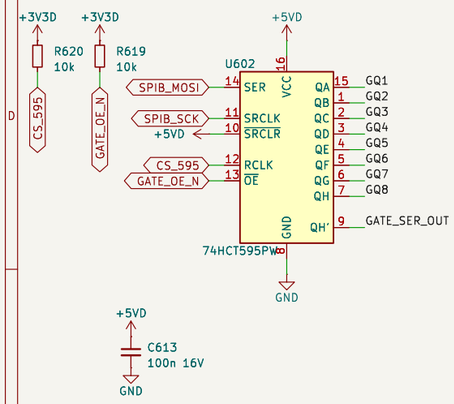
Panel bus test pads





Gate expansion: QH', SRCLK, RCLK, OE

Chain a second 74HCT595 off-board from J603 (74HCT595 only: a 74HC595 at 5 V needs 3.15 V inputs). OE keeps it off at boot too.



SRCLR tied high. Boot order: OE off, clear the register, then enable (firmware).

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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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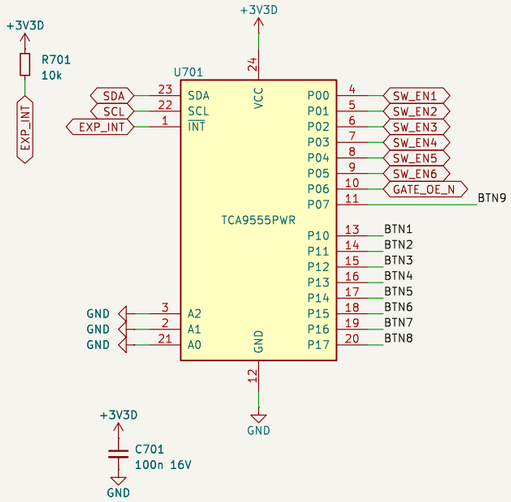
Sheet: /Pots & gate outs/
File: pots_gates.kicad_sch

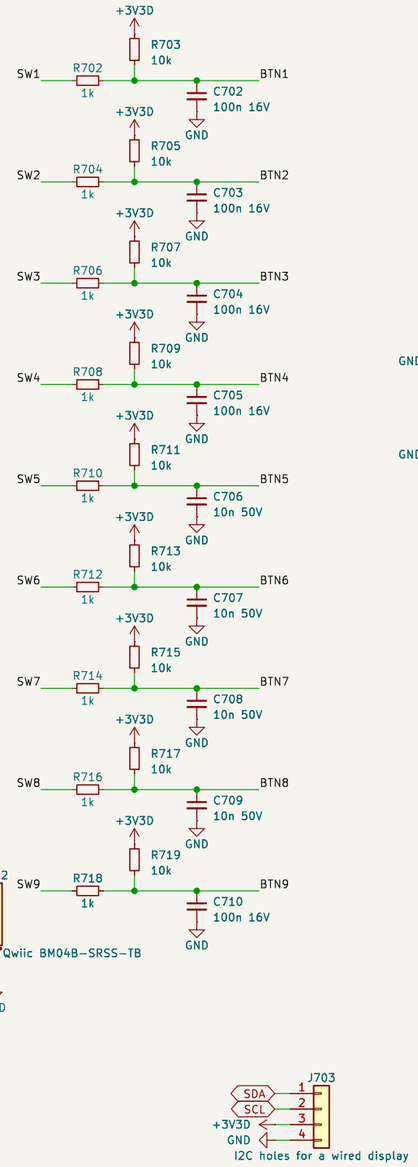
Title: Pots & gate outs

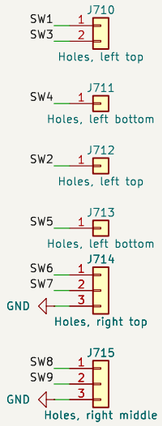
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 6/10

EXPANDER, I2C AND LEDS

TCA9555 at 0x20, M24C64 at 0x50; a Qwiic display is typically 0x3C/0x3D. 2.2 kΩ I2C pull-ups for 400 kHz.
The TCA9555 powers up with every line an input pulled high internally (~100 kΩ): switch enables and the gate
OE are defined by the pulls on their own sheets. Buttons and switches use P07 and P10–P17: hole → 1 kΩ → node with
10 kΩ to +3V3D and 100 nF (10 nF on SW5–SW8) → expander pin, as EuroPi does. A button pulls its hole to GND.
ENCODERS: wire A and B to gate inputs 3 and 4 (a clean comparator into a direct GPIO; encoder common to a +3V3D hole),
not to the expander: a busy OLED holds the shared I2C bus for ~23 ms and steps get lost. The push switch can use any SW hole.

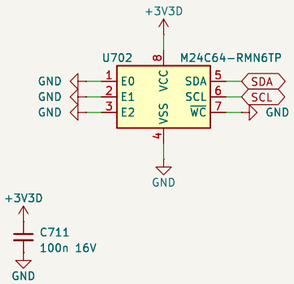




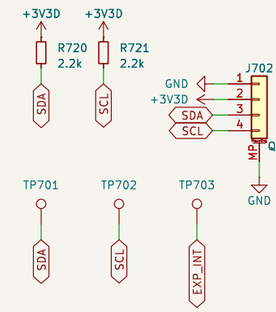


Encoder A/B: use gate inputs
3–4 (see note above); push
switch on any SW hole.

E0–E2 low: 0x50. WC low: writable.

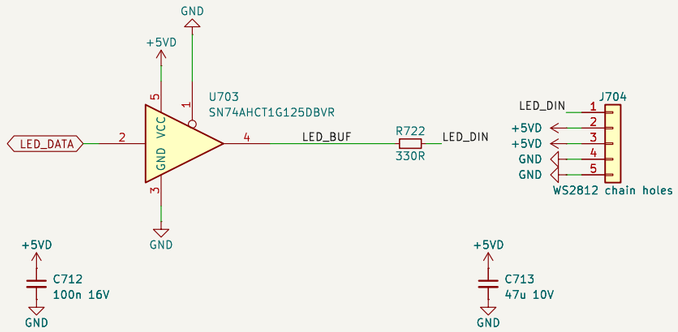


Qwiic: 1 GND, 2 3.3 V, 3 SDA, 4 SCL





WS2812 data: 3.3 V → 74AHCT1G125 at 5 V → 330 Ω → LED_DIN. LED power from the +5VD holes.



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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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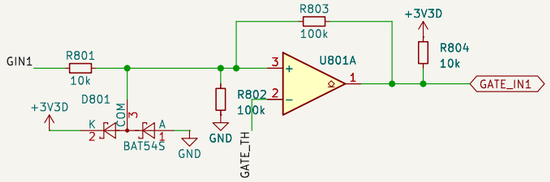
Sheet: /Expander, I2C & LEDs/
File: expander_i2c.kicad_sch

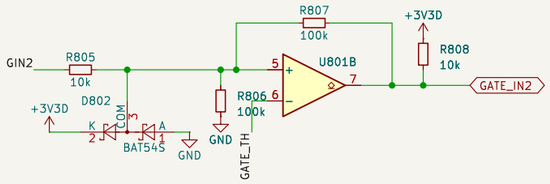
Title: Expander, I2C & LEDs

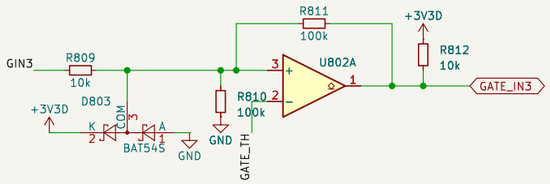
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 7/10

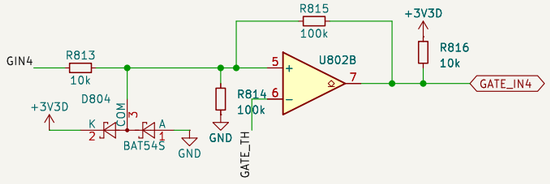
GATE INPUTS AND JACK DETECT

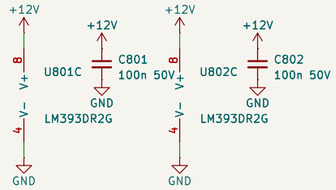
Adapted from Ogham's clock input (Ogham feeds the comparator's − input; this one is non-inverting, so four channels share one threshold divider): 10 kΩ, BAT54S clamps to +3V3D / GND, 100 kΩ to ground, LM393 against GATE_TH = 1.65 V with 100 kΩ hysteresis. Trips at ≈1.98 V rising, ≈1.67 V falling at the jack. The LM393 runs from +12 V so the clamped input (up to ≈3.6 V) stays inside its common-mode range (VCC − 1.5 V; at 5 V that would be 3.5 V). Open-collector output pulled up to +3V3D: GATE_INn is high while the gate is high, safe for any module. Firmware: catch 100 µs triggers with edge interrupts or PIO; mask the inputs while the jack-detect probe runs. The clamps push up to ≈3.4 mA into +3V3D: the 680 Ω minimum load on the power sheet absorbs it.



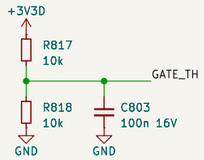




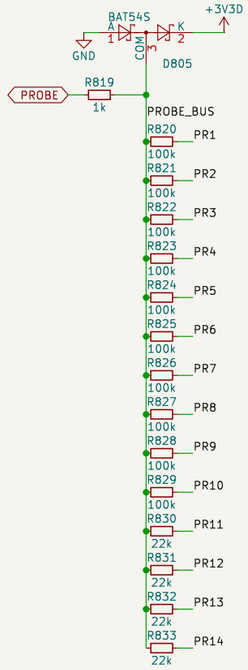


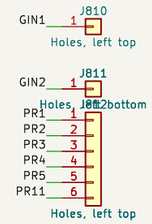


GATE_TH = +3V3D / 2



JACK DETECT (Workshop System Computer): PROBE drives a pseudo-random bit through a resistor to each input's switch (normal) lug. Unpatched, the jack follows the sequence; patched, the lug opens. 100 kΩ for CV and audio inputs; 22 kΩ for gate inputs, whose 1.98 V threshold a 100 kΩ probe cannot reach (it gives ≈1.24 V).





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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md

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Sheet: /Gate ins & jack detect/

File: gate_in.kicad_sch

Title: Gate ins & jack detect

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 8/10

EDGE STRIPS

Two n8synth-style edge strips, 40 positions each at 2.54 mm. Each position is an outer hole (the deck's header pin) joined to an inner hole (a wiring point): both pads carry the same pad number, so they share a net. Nothing on the board connects to them: each module is wired by hand from the channel holes to the inner holes. Deck cells occupy positions 1–10, 16–25 and 31–40; 11–15 and 26–30 are free tie points. 10HP: the board sits parallel behind the deck and both strips are used. 6HP: the board sits at 90° and only the left strip meets the deck's right-angle header.

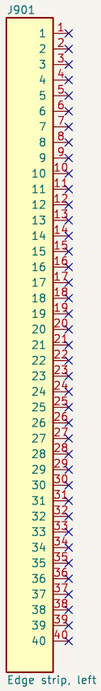
Fiducials for JLCPCB assembly

FID901 Fiducial

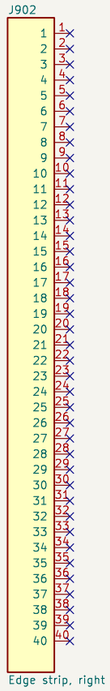
FID902 Fiducial

FID903 Fiducial

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Edge strip, left



Edge strip, right

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Sheet: /Edge strips/
File: edge_strips.kicad_sch

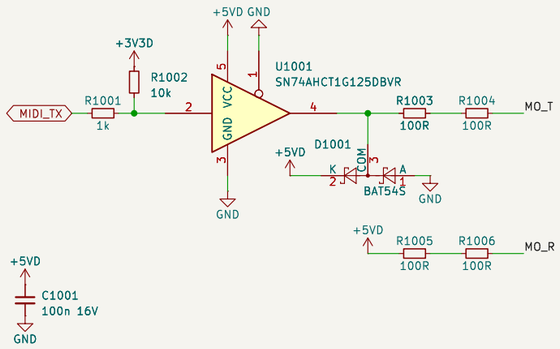
Title: Edge strips

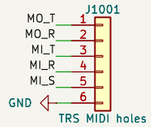
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 9/10

TRS MIDI IN AND OUT

MIDI 1.0 over 3.5 mm TRS, Type A (MIDI Association RP-054): tip = current sink (DIN pin 5), ring = current source
(DIN pin 4), sleeve = shield (DIN pin 2). Electrical values per CA-033 (2014). Uses the two UART-capable spare pins through
link holes on J101 (SPARE1-MIDI_TX, SPARE2-MIDI_RX): left open, SPARE1/2 stay plain GPIO. Review: docs/analogue_board_review/midi.md

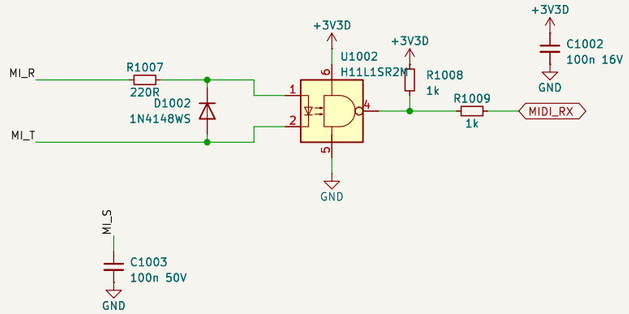
MIDI OUT: 3.3 V UART → 74AHCT1G125 at +5VD → 2 × 100 Ω per leg (1206) → TRS. Idle high = no loop current.
A mono cable shorting ring to sleeve draws 25 mA from the buck; each 100 Ω sees 64 mW.





WIRING (two stereo jacks in deck cells):
OUT: tip → MO_T, ring → MO_R, sleeve → the deck's D (ground) pad.
IN: tip → MI_T, ring → MI_R, sleeve → MI_S or open: NOT to D.
Turn MIDI on: link SPARE1-MIDI_TX and SPARE2-MIDI_RX on J101.
Type B gear: swap that jack's tip and ring wires.
Bring-up: loop OUT to IN with a stereo cable: ≈1.2-1.3 V
across the 220 Ω while sending = 5.5-5.9 mA in the loop.
The H11L1 is ≈4 mm tall: place it outside the module footprint.

MIDI IN: ring → 220 Ω → H11L1 LED → tip; 1N4148WS across the LED for reverse polarity. H11L1 switches at 1.6 mA
(a worst-case 3.3 V sender still gives ≈4.5 mA). Output pulled up by 1 kΩ to +3V3D, then 1 kΩ to the link hole.
The IN jack's sleeve is NOT grounded (CA-033: no ground loop): wire it to MI_S (100 nF to ground) or leave it open.



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Checkpoint 1 draft: schematic for review. Spec: docs/analogue_board_spec.md
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Sheet: /TRS MIDI/		
File: midi.kicad_sch		
Title: TRS MIDI		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 10/10